PROMPT LIFE V0.8.0

Lesson Cards

One accessible curriculum-review card per lesson, built from current app data plus inventory notes and rubric scores.

What Is an LLM?

A learned next-token prediction system

Stage: architecture | Priority: low | Rubric avg: 4.4/5

DEFINITION

A large language model is a learned system that turns current context into probabilities for the next token.

CORE EXPLANATION

An LLM does not read a prompt and write a whole answer at once. During inference, the current context enters the model, learned weights shape internal hidden states, the model scores possible next tokens, one token is chosen, and that token is appended before the next run. Training built the weights earlier; inference uses them.

WHERE IT HAPPENS

This card describes the whole model lifecycle, but especially normal inference: current context to next-token probabilities to generated response token.

DURABLE VS TEMPORARY

The weights were changed during training or fine-tuning. During ordinary inference, hidden states are temporary and weights normally stay fixed.

PROMPT VS RESPONSE

Prompt tokens are given to the model. Response tokens are generated one at a time.

WHY IT MATTERS

This definition lowers the mystery. The model can be powerful without being a mind, database, or hand-written rulebook.

RELATIONSHIP

Next, we compare the older rule-based dream of AI with the deep-learning approach that made LLMs work.

METAPHOR

A vast autocomplete engine with learned structure.

BRAIN METAPHOR

Like a person using context to anticipate what word might come next.

BRAIN LIMIT

A person has a body, goals, lived experience, and awareness. The model has learned weights and temporary computations.

MISCONCEPTION

An LLM is a conscious reader or a database.

VISUAL AID

`llm-overview`

CURRENT EXERCISE

`pick-next-token`: Pick the Next Token

CHECKPOINT

Which description is most accurate? Correct: A learned prediction system. Incorrect: A conscious reader; A hand-written rulebook; A search engine that knows every file.

FEEDBACK

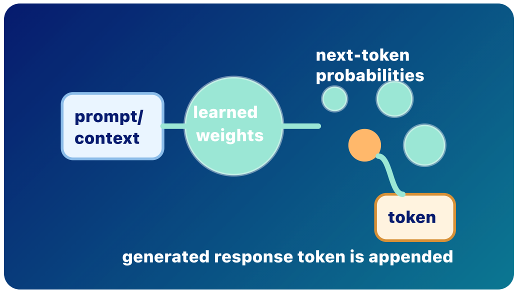
An LLM uses learned weights to turn context into next-token probabilities. Choice notes: A conscious reader: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook. A hand-written rulebook: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook. A search engine that knows every file: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.

REWRITE NOTES

Risk: An LLM is a conscious reader or a database.
Missing: An LLM does not read a prompt and write a whole answer at once. During inference, the current context enters the model, learned weights shape internal hidden states, the model scores possible next tokens, one token is chosen, and that token is appended before the next run. Training built the weights earlier; inference uses them.

ILLUSTRATION NEEDED

Current visual aid: llm-overview. One-page prompt lifecycle: prompt enters, fixed weights process, next token sampled, response grows.



Prompt to Prediction

Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.

Rubric Scores

Accuracy 5/5	**Placement** 4/5	**Relationship** 5/5	**Prompt/response** 5/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

llm | prompt | response | token | weight | inference

Rationalists vs Empiricists

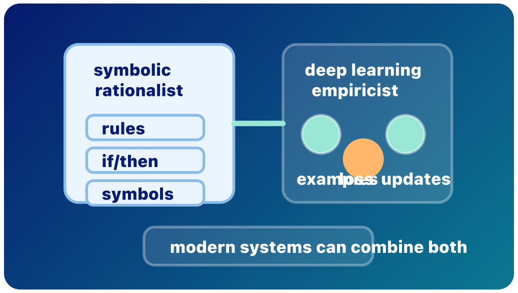
Rules versus learned patterns

Stage: architecture | Priority: medium | Rubric avg: 3.6/5

DEFINITION
Symbolic AI tries to reason with explicit rules and symbols; deep learning learns useful patterns from data by adjusting weights.

CORE EXPLANATION
Early AI often followed the rationalist dream: intelligence as explicit rules, logic, symbols, and search. Deep learning follows a more empiricist path: expose a model to many examples, measure error, and let the system learn internal representations that work. LLMs mostly belong to the deep-learning tradition, although real systems often combine learned models with rules, tools, retrieval, filters, and policies.

WHERE IT HAPPENS
This is historical and architectural background. It explains what kind of system an LLM is.



Rules and Learned Patterns
Symbolic/rationalist systems use explicit rules; deep-learning/empiricist systems learn useful patterns from examples.

DURABLE VS TEMPORARY
Deep learning changes weights during training. Symbolic rules are usually written or configured explicitly.

PROMPT VS RESPONSE
A prompt is not matched against one giant rulebook. It is processed through learned parameters that shape response-token probabilities.

WHY IT MATTERS
This helps learners stop expecting LLMs to behave like ordinary hand-written programs.

RELATIONSHIP
If LLMs are learned systems, then the next question is: how are they learned? That leads to training.

METAPHOR
A rulebook versus a trained landscape.

BRAIN METAPHOR
Humans use both explicit reasoning and learned intuition.

BRAIN LIMIT
An LLM is not a full human mind combining logic, experience, emotion, and embodiment. It is a learned prediction system.

MISCONCEPTION
LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

VISUAL AID
`traditions`

CURRENT EXERCISE
`durable-or-temporary`: Durable or Temporary?

CHECKPOINT
Which phrase best describes the LLM tradition? Correct: Learned patterns from data. Incorrect: Only fixed if-then rules; A spreadsheet macro; A conscious reasoning engine.

FEEDBACK
LLMs are learned prediction systems shaped by data. Choice notes: Only fixed if-then rules: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights. A spreadsheet macro: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights. A conscious reasoning engine: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.

REWRITE NOTES
Risk: LLMs are hand-coded rulebooks, or symbolic AI is obsolete. **Missing:** Early AI often followed the rationalist dream: intelligence as explicit rules, logic, symbols, and search. Deep learning follows a more empiricist path: expose a model to many examples, measure error, and let the system learn internal representations that work. LLMs mostly belong to the deep-learning tradition, although real systems often combine learned models with rules, tools, retrieval, filters, and policies.

ILLUSTRATION NEEDED
Current visual aid: traditions. Split-panel rulebook versus learned landscape, with a small note that systems can combine them.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

rationalism · empiricism · symbolic-ai · deep-learning · training · weight

Training

Prediction error updates weights

Stage: pretraining | Priority: low | Rubric avg: 4.3/5

DEFINITION
Training is the process that adjusts model weights by comparing predictions with target answers, such as actual next tokens.

CORE EXPLANATION
During training, the model sees examples, predicts something, measures how wrong it was, and updates its weights to do slightly better next time. This happens over many examples and many updates. Training is where durable model capability is created.

WHERE IT HAPPENS
Before ordinary use, in a separate optimization process.

DURABLE VS TEMPORARY
Training durably changes weights. Ordinary inference does not.

PROMPT VS RESPONSE
Training examples may look like text prompts and completions, but the purpose is different: the system is updating weights, not merely producing a live response for a user.

WHY IT MATTERS
This is the dividing line between learning and using. Many AI myths come from confusing training with inference.

RELATIONSHIP
Pretraining is the broad first version of this training loop.

METAPHOR
Practice that changes the instrument before the performance.

BRAIN METAPHOR
Like practice shaping future habits or skill.

BRAIN LIMIT
The model is not reflecting on mistakes. An optimizer changes numbers to reduce error.

MISCONCEPTION
When I chat with the model, it automatically trains itself.

VISUAL AID
`training-loop`

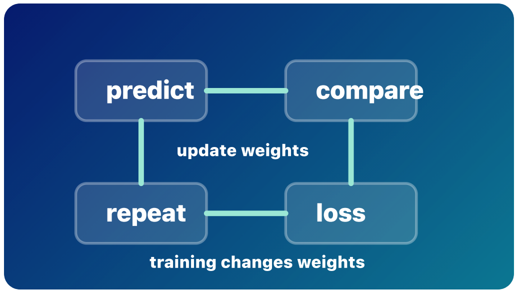
CURRENT EXERCISE
`training-nudge`: Training Nudge

CHECKPOINT
What makes training different from inference? Correct: Training durably updates weights. Incorrect: Training only reads the current context window; Training is the same as sampling; Training writes the final answer.

FEEDBACK
Training changes the model; inference uses the model. Choice notes: Training only reads the current context window: Not quite. Training is a weight-changing process, not ordinary response generation. Training is the same as sampling: Not quite. Training is a weight-changing process, not ordinary response generation. Training writes the final answer: Not quite. Training is a weight-changing process, not ordinary response generation.

REWRITE NOTES
Risk: When I chat with the model, it automatically trains itself. **Missing:** During training, the model sees examples, predicts something, measures how wrong it was, and updates its weights to do slightly better next time. This happens over many examples and many updates. Training is where durable model capability is created.

ILLUSTRATION NEEDED
Current visual aid: training-loop. Training loop with predict, compare, loss, update weights, repeat.



Training Loop

Predict, compare, loss, update weights, repeat. Training changes weights.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

training | loss | weight | weight-update | parameter | training-data | inference | forward pass

Pretraining

Broad learning before normal use

Stage: pretraining | Priority: low | Rubric avg: 4.1/5

DEFINITION

Pretraining is the broad first training phase where a model learns general patterns by repeatedly predicting tokens across large datasets.

CORE EXPLANATION

Before a model can be useful in ordinary conversation, it is pretrained on large collections of text and other data. The model repeatedly predicts tokens, measures error, and updates weights. Over time it learns grammar, styles, facts, associations, reasoning patterns, and task shapes. This does not mean it stores every source as a perfect memory.

WHERE IT HAPPENS

Before deployment, during the broad foundation-building stage.

DURABLE VS TEMPORARY

Pretraining durably changes weights.

PROMPT VS RESPONSE

Pretraining creates the weights later used to process prompts and generate responses.

WHY IT MATTERS

Pretraining is why next-token prediction can become surprisingly capable. It installs broad pattern knowledge into weights.

RELATIONSHIP

Pretraining builds broad capability. Overfitting explains a failure mode: learning examples too narrowly instead of patterns that generalize.

METAPHOR

Billions of tiny raindrops carving paths through a landscape.

BRAIN METAPHOR

Like broad education shaping later intuition.

BRAIN LIMIT

Humans connect learning to lived experience. A model stores statistical structure in parameters.

MISCONCEPTION

Pretraining means the model can perfectly recall everything it saw.

VISUAL AID

`pretraining-rain`

CURRENT EXERCISE

`training-nudge`: Training Nudge

CHECKPOINT

During pretraining, what changes? Correct: Model weights. Incorrect: Only the current chat; Only the user interface; Nothing changes.

FEEDBACK

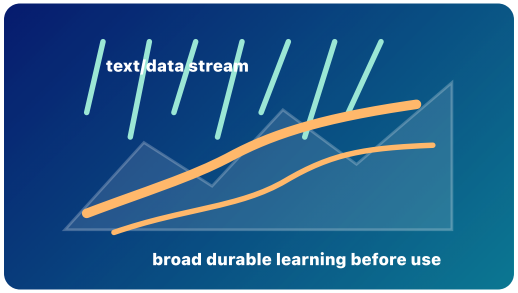
Pretraining changes weights before normal use. Choice notes: Only the current chat: Not quite. Pretraining is a durable training process, not a temporary chat event. Only the user interface: Not quite. Pretraining is a durable training process, not a temporary chat event. Nothing changes: Not quite. Pretraining is a durable training process, not a temporary chat event.

REWRITE NOTES

Risk: Pretraining means the model can perfectly recall everything it saw. **Missing:** Before a model can be useful in ordinary conversation, it is pretrained on large collections of text and other data. The model repeatedly predicts tokens, measures error, and updates weights. Over time it learns grammar, styles, facts, associations, reasoning patterns, and task shapes. This does not mean it stores every source as a perfect memory.

ILLUSTRATION NEEDED

Current visual aid: pretraining-rain. Broad data rain carving durable pathways into a model landscape.



Broad Pretraining

A wide stream of data shapes broad durable learning before normal use.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 2/5	**Mobile** 4/5

pretraining · training · training-data · weight · loss · next-token · inference

Overfitting and Generalization

Memorizing examples versus learning patterns

Stage: pretraining | Priority: high | Rubric avg: 4.4/5

DEFINITION

Overfitting happens when a model learns training examples too narrowly and performs poorly on new examples; generalization means learned patterns work beyond the examples seen during training.

CORE EXPLANATION

A model can fit old examples very well but still fail on new cases. That is overfitting. Good learning captures patterns that transfer. For LLMs, generalization is why a model can answer prompts it has never seen exactly before. Overfitting is why evaluation needs unseen examples, not just training performance.

WHERE IT HAPPENS

During training, evaluation, and model selection.

DURABLE VS TEMPORARY

Overfitting is about what gets installed into weights during training. It is not a temporary inference effect.

PROMPT VS RESPONSE

A model that generalizes well can handle new prompts. A model that overfits may repeat brittle patterns from training.

WHY IT MATTERS

This helps learners see why more memorization is not the same as better intelligence.

RELATIONSHIP

Pretraining should build broad patterns. Fine-tuning must adapt behavior without making the model too narrow.

METAPHOR

Studying only the answer key versus learning the subject.

BRAIN METAPHOR

Like memorizing practice questions without understanding the principle.

BRAIN LIMIT

The model is not consciously memorizing. Optimization can make it rely on brittle numerical patterns.

MISCONCEPTION

If a model performs well on training data, it has learned well.

VISUAL AID

`overfitting-generalization`

CURRENT EXERCISE

`durable-or-temporary`: Durable or Temporary?

CHECKPOINT

Which is a sign of overfitting? Correct: Great performance on training examples but poor performance on new examples. Incorrect: A model using a context window; A model retrieving documents with RAG; A model generating one token at a time.

FEEDBACK

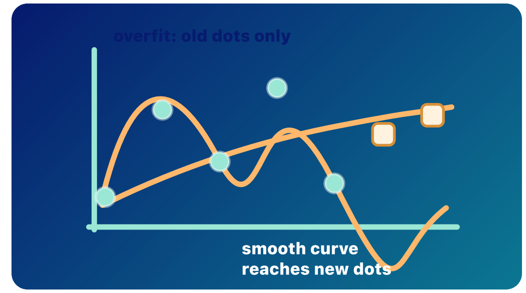
Generalization matters more than memorizing the answer key. Choice notes: A model using a context window: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval. A model retrieving documents with RAG: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval. A model generating one token at a time: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.

REWRITE NOTES

Risk: If a model performs well on training data, it has learned well. **Missing:** A model can fit old examples very well but still fail on new cases. That is overfitting. Good learning captures patterns that transfer. For LLMs, generalization is why a model can answer prompts it has never seen exactly before. Overfitting is why evaluation needs unseen examples, not just training performance.

ILLUSTRATION NEEDED

Current visual aid: overfitting-generalization. Two curves: an overfit curve that traces training dots too tightly and a smoother curve that predicts new dots.



Overfitting vs Generalization

Memorizing examples is not the same as learning transferable patterns.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 5/5	Exercise 4/5	Mobile 4/5

overfitting · generalization · validation-data · training-data · evaluation · training

Fine-Tuning

Targeted shaping after pretraining

Stage: fine-tuning | Priority: low | Rubric avg: 4.1/5

DEFINITION

Fine-tuning is additional training that adapts a pretrained model toward a task, domain, style, or preference.

CORE EXPLANATION

A pretrained model knows broad language patterns, but it may not behave the way users need. Fine-tuning adds targeted examples or preference data so future responses are more helpful, specialized, safer, or better aligned with a task. Fine-tuning can update model weights directly or add smaller adapter weights.

WHERE IT HAPPENS

After pretraining and before or between deployment cycles.

DURABLE VS TEMPORARY

Fine-tuning can durably change weights or add durable adapter behavior. It is different from a prompt, which usually only steers the current run.

PROMPT VS RESPONSE

Fine-tuning shapes future response patterns. It does not generate the current response token by token.

WHY IT MATTERS

This explains how one base model can become many differently behaving assistants.

RELATIONSHIP

Fine-tuning adapts a model; alignment is one major reason to fine-tune or otherwise shape behavior.

METAPHOR

Carving useful trails through an already huge terrain.

BRAIN METAPHOR

Like coaching someone to follow a house style or professional norm.

BRAIN LIMIT

The model is not choosing a new identity or values. Training changes output patterns.

MISCONCEPTION

Prompting once is the same as fine-tuning.

VISUAL AID

`fine-tune-path`

CURRENT EXERCISE

`durable-or-temporary`: Durable or Temporary?

CHECKPOINT

Which action is closest to fine-tuning? Correct: Additional training that shapes future responses. Incorrect: Typing one better prompt; Retrieving one PDF into context; Sampling the next token.

FEEDBACK

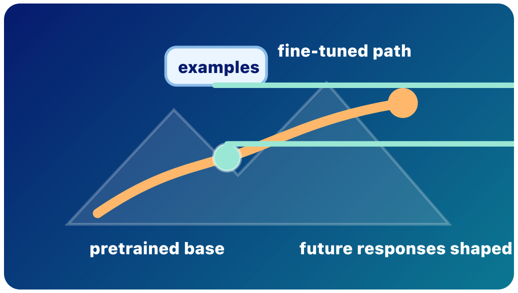
Fine-tuning changes future behavior more durably than a single prompt. Choice notes: Typing one better prompt: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training. Retrieving one PDF into context: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training. Sampling the next token: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.

REWRITE NOTES

Risk: Prompting once is the same as fine-tuning.
Missing: A pretrained model knows broad language patterns, but it may not behave the way users need. Fine-tuning adds targeted examples or preference data so future responses are more helpful, specialized, safer, or better aligned with a task. Fine-tuning can update model weights directly or add smaller adapter weights.

ILLUSTRATION NEEDED

Current visual aid: fine-tune-path. Targeted trail added onto an existing pretrained terrain.



Fine-Tuning Path

Targeted examples nudge an already-trained model toward a desired pattern.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 2/5	**Mobile** 4/5

fine-tuning | adapter | pretraining | training | prompt | response | alignment

Alignment

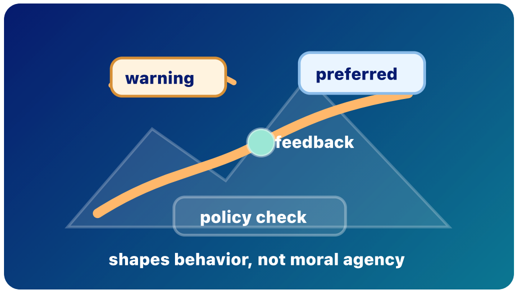
Shaping behavior toward human goals

Stage: alignment | Priority: high | Rubric avg: 4.5/5

DEFINITION
Alignment is the effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values.

CORE EXPLANATION
A model trained only to predict likely text may produce outputs that are fluent but unhelpful, unsafe, biased, or misleading. Alignment uses methods such as instruction tuning, human feedback, preference optimization, policies, filters, evaluations, and system design to shape model behavior. Alignment is not one switch. It is an ongoing effort to make model behavior more useful, reliable, and compatible with human goals.

WHERE IT HAPPENS
During fine-tuning, human feedback learning, system design, evaluation, deployment, and policy enforcement.



Alignment Landscape

Alignment shapes behavior with preferred paths, guardrails, feedback, and policies; it does not create moral agency.

DURABLE VS TEMPORARY
Some alignment work durably changes weights or adapter weights. Some alignment happens through system prompts, filters, retrieval, tool rules, or runtime safeguards that steer behavior temporarily.

PROMPT VS RESPONSE
Alignment affects how the model responds to prompts, but it does not mean the model understands morality like a person.

WHY IT MATTERS
Alignment is why a model may refuse harmful requests, follow instructions, or prefer clearer helpful answers over merely likely text.

RELATIONSHIP
Fine-tuning can shape model behavior. Alignment asks: toward what goals and constraints should that behavior be shaped?

METAPHOR
Guardrails, coaching, and a compass.

BRAIN METAPHOR
Like learning social norms or professional expectations.

BRAIN LIMIT
The model does not acquire human values, moral agency, conscience, or care. Its behavior is shaped by training and system constraints.

MISCONCEPTION
Aligned means morally good or fully trustworthy.

VISUAL AID
`alignment`

CURRENT EXERCISE
`durable-or-temporary`: Durable or Temporary?

CHECKPOINT
What does alignment try to do? Correct: Shape model behavior toward human goals, instructions, and safety boundaries. Incorrect: Make the model conscious; Guarantee every answer is true; Permanently solve all AI risk.

FEEDBACK
Alignment shapes behavior, but it is not magic morality. Choice notes: Make the model conscious: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee. Guarantee every answer is true: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee. Permanently solve all AI risk: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

REWRITE NOTES
Risk: Aligned means morally good or fully trustworthy.
Missing: A model trained only to predict likely text may produce outputs that are fluent but unhelpful, unsafe, biased, or misleading. Alignment uses methods such as instruction tuning, human feedback, preference optimization, policies, filters, evaluations, and system design to shape model behavior. Alignment is not one switch. It is an ongoing effort to make model behavior more useful, reliable, and compatible with human goals.

ILLUSTRATION NEEDED
Current visual aid: alignment. Response landscape with preferred paths, warning zones, guardrails, policy checkpoint, and human feedback marker.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 5/5	**Visual** 5/5	**Exercise** 4/5	**Mobile** 4/5

alignment · instruction-tuning · human-feedback-learning · rlhf · preference-optimization · policy · guardrail · evaluation · fine-tuning

Inference

A forward pass, not a memory update

Stage: inference | Priority: low | Rubric avg: 4.4/5

DEFINITION

Inference is normal model use: fixed weights process the current context to generate response tokens without durable weight updates.

CORE EXPLANATION

Inference is normal model use: fixed weights process the current context to generate response tokens without durable weight updates.

WHERE IT HAPPENS

It happens every time a deployed model answers a prompt.

DURABLE VS TEMPORARY

Not listed in current app data.

PROMPT VS RESPONSE

This is the live prompt-to-response path: fixed weights process current context and generate response tokens.

WHY IT MATTERS

This is the line between using a trained model and training a model.

RELATIONSHIP

Each forward pass turns the current context into next-token scores; decoding chooses one generated token for the response.

METAPHOR

Using a map to choose a route, not redrawing the map.

BRAIN METAPHOR

It can feel like thinking because internal representations change temporarily.

BRAIN LIMIT

Those hidden states are temporary computations, not memories, beliefs, or learning.

MISCONCEPTION

Temporary hidden states may be mistaken for new memories or weight changes.

VISUAL AID

`inference-pass`

CURRENT EXERCISE

`training-nudge`: Training Nudge

CHECKPOINT

What changes during ordinary inference? Correct: Temporary hidden states. Incorrect: The model weights permanently; The training dataset.

FEEDBACK

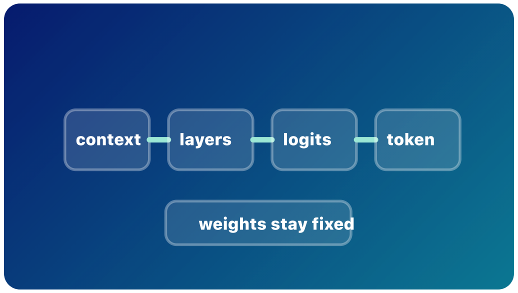
Inference creates temporary internal vectors for this run, then discards or replaces them.

REWRITE NOTES

Risk: Temporary hidden states may be mistaken for new memories or weight changes. **Missing:** Make the forward pass tangible: fixed weights plus temporary activations produce one set of next-token scores.

ILLUSTRATION NEEDED

Current visual aid: inference-pass. Forward-pass pipeline with fixed weights underneath and temporary states above.



Forward Pass

Inference creates temporary states while durable weights stay fixed.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 5/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

inference · forward pass · prompt · response · hidden state · logits

Prompt vs Response

Given context versus generated tokens

Stage: prompt processing | Priority: low | Rubric avg: 4.4/5

DEFINITION
The prompt is the input context supplied to the model; the response is the sequence of tokens the model generates after reading it.

DURABLE VS TEMPORARY
Not listed in current app data.

RELATIONSHIP
Each generated response token is appended to the context, so it can influence the next forward pass.

BRAIN LIMIT
The model is not deciding what it means to say; it samples from learned next-token probabilities.

CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

REWRITE NOTES
Risk: Learners may think the whole answer appears at once instead of being generated token by token.
Missing: Add the response-so-far as a visible part of the next input context.

CORE EXPLANATION
The prompt is the input context supplied to the model; the response is the sequence of tokens the model generates after reading it.

PROMPT VS RESPONSE
This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

METAPHOR
Given cards on a table, then new cards added one at a time.

MISCONCEPTION
Learners may think the whole answer appears at once instead of being generated token by token.

CHECKPOINT
What happens after one response token is generated? Correct: It is appended to the context. Incorrect: The whole answer appears at once; The model permanently learns it.

ILLUSTRATION NEEDED
Current visual aid: prompt-response. Given prompt cards and newly generated response cards entering the next context together.

WHERE IT HAPPENS
Prompt tokens are present before the next token is chosen; response tokens are added one at a time as generation proceeds.

WHY IT MATTERS
The distinction prevents a common myth that the model writes a full answer all at once.

BRAIN METAPHOR
It resembles a person continuing a sentence after reading a prompt.

VISUAL AID
`prompt-response`

FEEDBACK
The response grows by next token, append, repeat.

Prompt vs Response
Prompt tokens are given; response tokens are generated and appended.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

prompt | response | prompt tokens | response tokens | generated token | context window

Tokenization

Text becomes chunks

Stage: prompt processing | Priority: medium | Rubric avg: 4.2/5

DEFINITION
Tokenization splits prompts and generated responses into chunks the model can represent as token IDs.

CORE EXPLANATION
Tokenization splits prompts and generated responses into chunks the model can represent as token IDs.

WHERE IT HAPPENS
It happens before embedding lookup and before the transformer layers process the input.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

WHY IT MATTERS
Models do not carry raw English sentences through their layers; they carry token IDs and numerical vectors.

RELATIONSHIP
Prompt tokens and earlier response tokens in the input context become token IDs, embeddings, tensors, and temporary hidden states.

METAPHOR
Parcels on a conveyor belt entering the model.

BRAIN METAPHOR
It loosely resembles hearing a sentence as parts before interpreting it.

BRAIN LIMIT
The parts are tokenizer chunks, not human phonemes, concepts, or meanings.

MISCONCEPTION
Tokens may be mistaken for whole words or human concepts.

VISUAL AID
`tokenization`

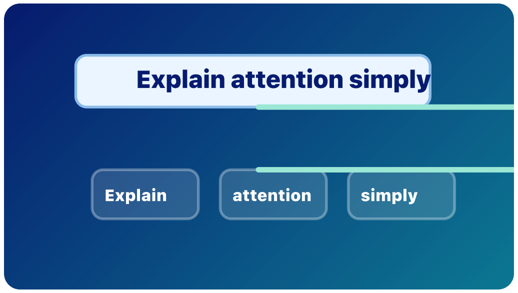
CURRENT EXERCISE
`prompt-or-response-label`: Prompt or Response?

CHECKPOINT
What comes right after text is split into tokens? Correct: Token IDs and embedding lookup. Incorrect: Softmax; A finished answer.

FEEDBACK
Prompt tokens and generated response tokens both map to IDs, and IDs retrieve learned embedding vectors.

REWRITE NOTES
Risk: Tokens may be mistaken for whole words or human concepts. **Missing:** Show that token boundaries can be word pieces, punctuation, or spaces, not necessarily words.

ILLUSTRATION NEEDED
Current visual aid: tokenization. Tokenizer conveyor splitting a short sentence into uneven chunks.



Text to Tokens

Text is split into model-readable chunks before embedding lookup.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 5/5	**Durable/temp** 3/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

token · prompt tokens · response tokens · token-id · embedding

Token IDs

Chunks become lookup numbers

Stage: prompt processing **Priority: low** **Rubric avg: 4.1/5**

DEFINITION
A token ID is the number assigned to a token so the model can look up its learned starting vector.

CORE EXPLANATION
A token ID is the number assigned to a token so the model can look up its learned starting vector.

WHERE IT HAPPENS
It sits between tokenization and the embedding table.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

WHY IT MATTERS
The ID is the bridge from text chunks to the learned numerical space the model uses.

RELATIONSHIP
Prompt tokens and already-generated response tokens use token IDs when they are part of the current input context.

METAPHOR
A library call number that points to the right shelf in the embedding table.

BRAIN METAPHOR
The lookup step is faintly like recognizing a familiar symbol.

BRAIN LIMIT
The number itself is not understanding; it is a key into a learned table.

MISCONCEPTION
The ID can be mistaken for meaning rather than a lookup key.

VISUAL AID
`token-ids`

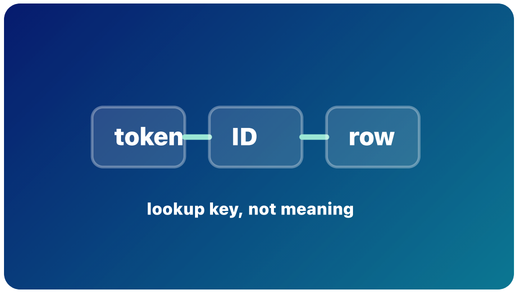
CURRENT EXERCISE
`prompt-or-response-label`: Prompt or Response?

CHECKPOINT
Why does the model need token IDs? Correct: To look up embeddings. Incorrect: To store a permanent memory; To skip tensors.

FEEDBACK
Token IDs connect text chunks to learned embedding vectors.

REWRITE NOTES
Risk: The ID can be mistaken for meaning rather than a lookup key. **Missing:** Make the tokenizer and embedding table two separate pieces in the diagram.

ILLUSTRATION NEEDED
Current visual aid: token-ids. Token chunks mapped to numeric IDs, then to rows in an embedding table.



Token IDs
Each token gets a lookup number that points to an embedding row.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 3/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

token-id **prompt tokens** **response tokens** **embedding**

Embeddings

Token IDs look up learned vectors

Stage: prompt processing | **Priority: low** | **Rubric avg: 4.3/5**

DEFINITION
An embedding is a token ID's learned starting vector.

CORE EXPLANATION
An embedding is a token ID's learned starting vector.

WHERE IT HAPPENS
It is retrieved from the embedding table at the start of model processing.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card should stay explicit that prompt tokens and response-so-far tokens enter the same current context before the next token is chosen.

WHY IT MATTERS
Embeddings give each token a learned numerical starting point before context reshapes it.

RELATIONSHIP
Prompt tokens and already-generated response tokens are looked up as embeddings when they are inside the current context.

METAPHOR
A token gets a starting address in a giant meaning cloud.

BRAIN METAPHOR
It loosely resembles a starting association a person might have with a word.

BRAIN LIMIT
An embedding is not a definition or memory; it is a learned vector row.

MISCONCEPTION
Embeddings can be confused with definitions, memories, or hidden states.

VISUAL AID
`embeddings`

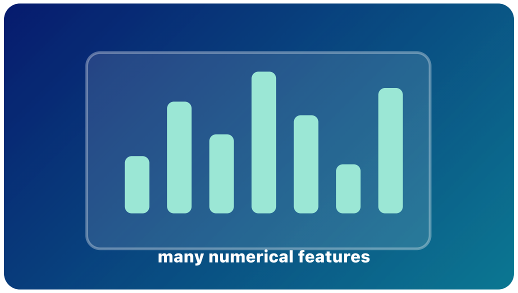
CURRENT EXERCISE
`prompt-or-response-label`: Prompt or Response?

CHECKPOINT
What is an embedding in this app? Correct: A learned starting vector. Incorrect: A finished sentence; A human memory.

FEEDBACK
A token ID retrieves an embedding vector before context reshapes it into hidden states.

REWRITE NOTES
Risk: Embeddings can be confused with definitions, memories, or hidden states. **Missing:** Contrast embedding as starting vector with hidden state as later context-shaped vector.

ILLUSTRATION NEEDED
Current visual aid: embeddings. Lookup table pulling a learned starting vector for one token ID.



Embedding Lookup
A token ID retrieves a learned starting vector.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 4/5	**Durable/temp** 5/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

embedding | **input context** | **token-id** | **vector** | **hidden state**

Vectors

Lists of numbers carrying features

Stage: architecture | Priority: high | Rubric avg: 3.8/5

DEFINITION
A vector is a list of numbers that represents learned features of something, such as a token.

CORE EXPLANATION
A vector is a list of numbers that represents learned features of something, such as a token.

WHERE IT HAPPENS
Embeddings and hidden states are both vectors at different moments in the model path.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
Vectors let the model compute with many fuzzy features at once instead of using one plain-English label.

RELATIONSHIP
An embedding vector starts the token; hidden-state vectors are later context-shaped versions of that token.

METAPHOR
A coordinate in a many-dimensional map.

BRAIN METAPHOR
It can resemble a bundle of associations around a word.

BRAIN LIMIT
The vector is not a conscious concept; it is a numerical representation useful for computation.

MISCONCEPTION
Feature dimensions may be treated as literal labeled meanings rather than learned numerical patterns.

VISUAL AID
`vectors`

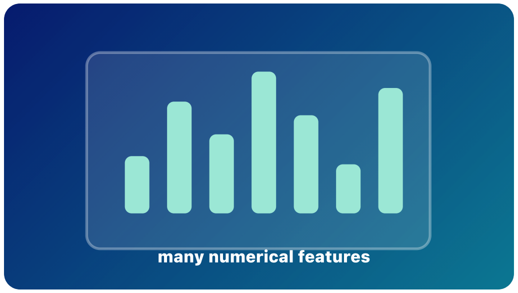
CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

CHECKPOINT
Which statement is most accurate? Correct: Embeddings and hidden states are vectors. Incorrect: Vectors are whole paragraphs; Vectors are permanent chats.

FEEDBACK
Vectors are lists of numbers. Embeddings are starting vectors; hidden states are temporary context-shaped vectors.

REWRITE NOTES
Risk: Feature dimensions may be treated as literal labeled meanings rather than learned numerical patterns. **Missing:** Explain that dimensions are learned and distributed; one feature is rarely a neat English label.

ILLUSTRATION NEEDED
Current visual aid: vectors. Many-dimensional feature bars with a warning that labels are simplified.



Feature Vector
A vector is a list of numerical features, not a sentence.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 3/5	Exercise 2/5	Mobile 4/5

vector | embedding | hidden state | feature

Tensors

Organized blocks of vectors

Stage: architecture | Priority: high | Rubric avg: 3.7/5

DEFINITION
A tensor is an organized block of numbers, often holding many token vectors at once.

CORE EXPLANATION
A tensor is an organized block of numbers, often holding many token vectors at once.

WHERE IT HAPPENS
Tensors carry the whole current context through transformer layers.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
A tensor keeps the model's data organized by positions and features, so many tokens can be processed together.

RELATIONSHIP
Embeddings from the current context become tensors so the transformer can process prompt tokens and earlier response tokens together.

METAPHOR
A stack of spreadsheets carrying token features through the model.

BRAIN METAPHOR
It loosely resembles organizing many signals at once.

BRAIN LIMIT
The tensor is not a mental image or thought; it is a shaped numerical array.

MISCONCEPTION
Tensor language can become jargon unless shapes and positions are made tangible.

VISUAL AID
`tensors`

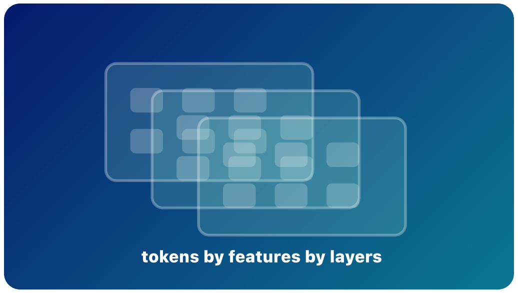
CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

CHECKPOINT
Which is the better definition of tensor here? Correct: An organized block of numbers. Incorrect: A word in a dictionary; A rule written by a programmer.

FEEDBACK
In transformers, tensors carry batches of token vectors through layers.

REWRITE NOTES
Risk: Tensor language can become jargon unless shapes and positions are made tangible. **Missing:** Show tensor axes: token position by feature, optionally across batches or heads only as a footnote.

ILLUSTRATION NEEDED
Current visual aid: tensors. Mobile-readable tensor block with token-position and feature axes.



Tensor Block
Tensors organize many token vectors for layer-by-layer processing.

Rubric Scores

Accuracy 5/5	**Placement** 4/5	**Relationship** 4/5	**Prompt/response** 3/5	**Durable/temp** 3/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 2/5	**Mobile** 4/5

tensor | vector | input context | layer

Attention

Weighted relevance between positions

Stage: architecture | Priority: high | Rubric avg: 4.4/5

DEFINITION
Attention assigns weighted relevance between token positions.

CORE EXPLANATION
Attention assigns weighted relevance between token positions.

WHERE IT HAPPENS
It happens inside transformer layers during a forward pass.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
Attention lets one token position use information from other visible token positions in the current context.

RELATIONSHIP
During inference, attention operates over token positions in the current context, including prompt tokens and earlier generated response tokens.

METAPHOR
Spotlights showing which earlier tokens are useful right now.

BRAIN METAPHOR
The word attention is useful because some parts of the context matter more than others.

BRAIN LIMIT
It is not human attention, awareness, desire, or focus. It is weighted relevance in a computation.

MISCONCEPTION
The term attention invites human-focus and awareness metaphors.

VISUAL AID
`attention`

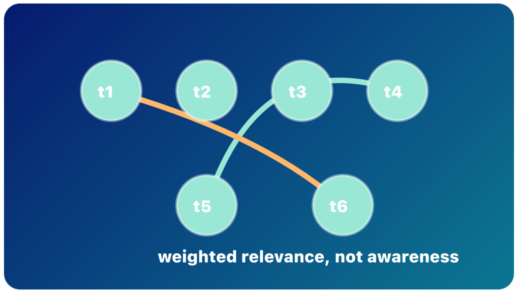
CURRENT EXERCISE
`attention-relevance`: Attention Is Relevance

CHECKPOINT
Which tokens can attention look across during inference? Correct: Tokens currently in the context window. Incorrect: All text the model has ever seen; Only the newest generated token.

FEEDBACK
Attention operates over the current context. It does not browse all training data.

REWRITE NOTES
Risk: The term attention invites human-focus and awareness metaphors. **Missing:** Show attention weights as arrows between positions, not as a mind selecting ideas.

ILLUSTRATION NEEDED
Current visual aid: attention. Token nodes with weighted relevance arcs and a plain-language caption.



Attention Weave
Attention is weighted relevance between token positions.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 5/5	**Durable/temp** 4/5
Metaphor 4/5	**Brain/cognition** 5/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

attention | input context | prompt tokens | response tokens | hidden state | layer

MLP

Feature reshaping per token

Stage: architecture | Priority: high | Rubric avg: 4.3/5

DEFINITION
An MLP is a feed-forward network that reshapes each token position's feature vector.

CORE EXPLANATION
An MLP is a feed-forward network that reshapes each token position's feature vector.

WHERE IT HAPPENS
It appears inside transformer layers, usually after attention has mixed information across positions.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
The MLP helps turn mixed context into more useful per-token features for later layers.

RELATIONSHIP
During a forward pass over the current context, attention mixes information across positions and the MLP reshapes features within each position.

METAPHOR
A forge that bends each token vector into a more useful shape.

BRAIN METAPHOR
It is somewhat like a small processing step after noticing relevant context.

BRAIN LIMIT
The MLP is not a neuron with feelings or beliefs; it is a learned function applied to vectors.

MISCONCEPTION
The MLP can blur into attention unless cross-position mixing and per-position reshaping are contrasted.

VISUAL AID
`mlp`

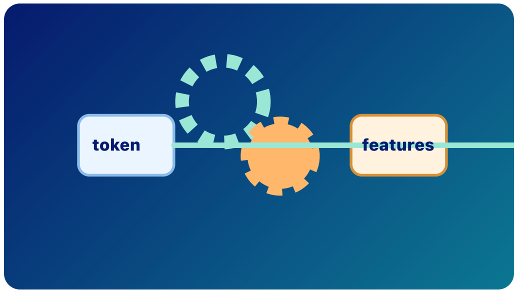
CURRENT EXERCISE
`mlp-feature-reshape`: MLP Feature Reshape

CHECKPOINT
What does the MLP mainly do here? Correct: Reshapes per-token features. Incorrect: Reads private files; Chooses the final answer directly.

FEEDBACK
MLPs transform token feature vectors after attention has mixed contextual information.

REWRITE NOTES
Risk: The MLP can blur into attention unless cross-position mixing and per-position reshaping are contrasted. **Missing:** Put MLP beside attention in the same layer so their complementary roles are visible.

ILLUSTRATION NEEDED
Current visual aid: mlp. Per-token vector forge or feature mixer after attention.



MLP Reshape
The MLP reshapes each token position feature vector.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 5/5	**Prompt/response** 4/5	**Durable/temp** 4/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

MLP | input context | hidden state | attention

Layers

Repeated refinement

Stage: architecture | Priority: medium | Rubric avg: 3.9/5

DEFINITION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

CORE EXPLANATION

A transformer layer is a repeated block that usually combines attention, MLP feature reshaping, residual paths, and normalization.

WHERE IT HAPPENS

Layers are stacked inside the model between embeddings and next-token scores.

DURABLE VS TEMPORARY

Not listed in current app data.

PROMPT VS RESPONSE

This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS

Repeated layers let simple starting vectors become richer, context-shaped hidden states.

RELATIONSHIP

Each layer refines hidden states while residual paths help useful information carry forward.

METAPHOR

A series of editing passes where each pass can revise but also preserve the draft.

BRAIN METAPHOR

The stack can feel like stages of processing.

BRAIN LIMIT

The layers do not form a human-like chain of thought; they update numerical states.

MISCONCEPTION

Layer stacks can be mistaken for a human chain of thought.

VISUAL AID

`layers`

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

CHECKPOINT

What do layers repeatedly refine? Correct: Hidden states. Incorrect: Permanent user memory; The app interface.

FEEDBACK

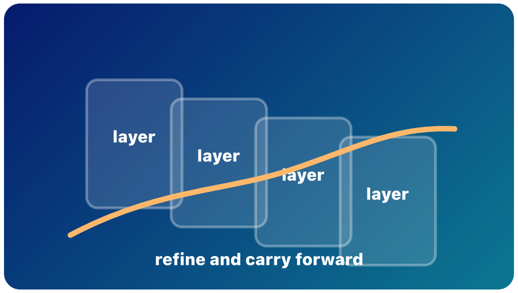
Layers update temporary internal vectors during inference.

REWRITE NOTES

Risk: Layer stacks can be mistaken for a human chain of thought. **Missing:** Add residual paths and normalization as optional advanced labels or a split target lesson.

ILLUSTRATION NEEDED

Current visual aid: layers. Transparent layer stack with attention and MLP blocks repeated.



Layer Stack

Repeated blocks refine hidden states while carrying useful signal forward.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

layer | attention | MLP | hidden state

Hidden States

Temporary contextual vectors

Stage: inference | Priority: high | Rubric avg: 4.4/5

DEFINITION
A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.

CORE EXPLANATION
A hidden state is the model's temporary internal context-shaped vector for a token at a given layer.

WHERE IT HAPPENS
Hidden states appear during the forward pass after embeddings enter the layer stack.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This is the live prompt-to-response path: fixed weights process current context and generate response tokens.

WHY IT MATTERS
They are where the prompt's context changes the token representation before the model scores possible next tokens.

RELATIONSHIP
Hidden states are created while processing the current context during a forward pass; they are not the visible response.

METAPHOR
A scratchpad of numbers that changes while the prompt is being processed.

BRAIN METAPHOR
Hidden state can resemble temporary working memory.

BRAIN LIMIT
It is not permanent memory, consciousness, or private English text; it is a temporary vector.

MISCONCEPTION
Hidden states can be mistaken for secret English text or permanent memory.

VISUAL AID
`hidden-states`

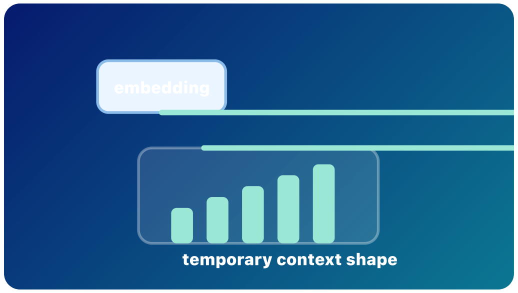
CURRENT EXERCISE
`mlp-feature-reshape`: MLP Feature Reshape

CHECKPOINT
Why is it called hidden state? Correct: It is internal numbers, not visible text. Incorrect: It is encrypted English; It is a secret memory file.

FEEDBACK
Hidden means inside the model's working representation.

REWRITE NOTES
Risk: Hidden states can be mistaken for secret English text or permanent memory. **Missing:** Show hidden state changing layer by layer for the same token in two contexts.

ILLUSTRATION NEEDED
Current visual aid: hidden-states. Same token glowing differently across contexts and layers.



Temporary Hidden State

Hidden states are context-shaped vectors created during a forward pass.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 4/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

hidden state · input context · forward pass · embedding · layer · inference

Logits

Raw next-token scores

Stage: response generation | Priority: medium | Rubric avg: 4.4/5

DEFINITION
Logits are raw scores for the next token before they become probabilities.

CORE EXPLANATION
Logits are raw scores for the next token before they become probabilities.

WHERE IT HAPPENS
They appear near the end of a forward pass, after the final hidden state is projected toward the vocabulary.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

WHY IT MATTERS
Logits show that the model has not chosen a token yet; it has scored candidates.

RELATIONSHIP
A forward pass over the current context produces logits for one next generated token.

METAPHOR
A scoreboard before the points are converted into odds.

BRAIN METAPHOR
It can feel like considering options.

BRAIN LIMIT
The model is not weighing options with intent; logits are raw numerical scores.

MISCONCEPTION
Raw scores can be confused with probabilities or truth confidence.

VISUAL AID
`logits`

CURRENT EXERCISE
`softmax-funnel`: Softmax Funnel

CHECKPOINT
Where do probabilities come from in this path? Correct: Softmax converts logits. Incorrect: Training rewrites the weights; The model asks a database.

FEEDBACK
Logits are raw scores; softmax turns them into probabilities.

REWRITE NOTES
Risk: Raw scores can be confused with probabilities or truth confidence. **Missing:** Make vocabulary projection explicit: final hidden state to one score per candidate token.

ILLUSTRATION NEEDED
Current visual aid: logits. Raw candidate scoreboard before softmax.

Raw Scores
Logits are raw next-token scores before probabilities.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 4/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

logits · generated token · forward pass · hidden state · softmax

Softmax

Scores become probabilities

Stage: response generation **Priority: medium** **Rubric avg: 4.4/5**

DEFINITION
Softmax turns next-token raw scores into probabilities that sum to one.

CORE EXPLANATION
Softmax turns next-token raw scores into probabilities that sum to one.

WHERE IT HAPPENS
It sits between logits and sampling during a decoding step.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

WHY IT MATTERS
Softmax makes scores comparable as a probability cloud over candidate tokens.

RELATIONSHIP
Softmax converts logits into probabilities; sampling then chooses one token from those probabilities.

METAPHOR
A funnel turning scoreboard points into odds.

BRAIN METAPHOR
It resembles turning vague preferences into relative chances.

BRAIN LIMIT
The probabilities are mathematical outputs, not confidence, desire, or truth.

MISCONCEPTION
Probabilities can be mistaken for correctness or factual confidence.

VISUAL AID
`softmax`

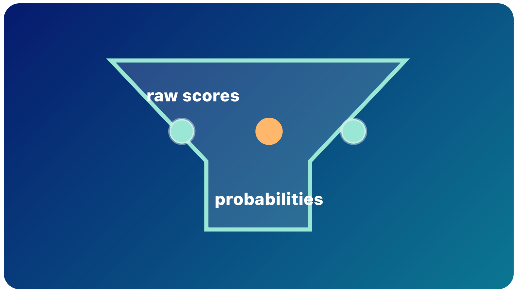
CURRENT EXERCISE
`softmax-funnel`: Softmax Funnel

CHECKPOINT
What does softmax produce? Correct: Next-token probabilities. Incorrect: Permanent memory; A whole essay.

FEEDBACK
Softmax turns raw next-token scores into a probability distribution.

REWRITE NOTES
Risk: Probabilities can be mistaken for correctness or factual confidence. **Missing:** Show that softmax normalizes scores into a distribution that sums to one.

ILLUSTRATION NEEDED
Current visual aid: softmax. Funnel converting raw scores into probabilities that sum to one.



Softmax Funnel
Softmax converts raw scores into probabilities that sum to one.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 5/5	**Prompt/response** 5/5	**Durable/temp** 4/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

softmax **logits** **decoding step** **sampling** **probability**

DECISION ROOM

Sampling

One next token is chosen

Stage: response generation | Priority: medium | Rubric avg: 4.3/5

DEFINITION
Sampling chooses one next token for the response from the probability cloud.

CORE EXPLANATION
Sampling chooses one next token for the response from the probability cloud.

WHERE IT HAPPENS
It happens after softmax in the decoding step.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

WHY IT MATTERS
Sampling explains why the same prompt can sometimes produce different good responses.

RELATIONSHIP
Temperature, top-k, and top-p shape the candidate pool before the chosen response token is appended to the context.

METAPHOR
Choosing from a weighted bowl of possible next tokens.

BRAIN METAPHOR
It can resemble choosing a word while speaking.

BRAIN LIMIT
The model is not choosing with intention; a decoding rule selects from probabilities.

MISCONCEPTION
Randomness can sound arbitrary unless tied to weighted probabilities.

VISUAL AID
`sampling`

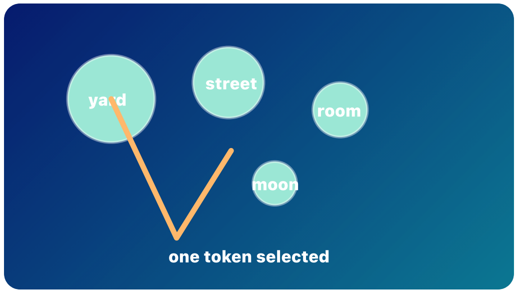
CURRENT EXERCISE
`pick-next-token`: Pick the Next Token

CHECKPOINT
When the model chooses one token, what happens next? Correct: The token is appended and the model runs again. Incorrect: The model writes the whole response at once; The model permanently learns the token.

FEEDBACK
The selected response token becomes part of the context for the next forward pass.

REWRITE NOTES
Risk: Randomness can sound arbitrary unless tied to weighted probabilities. **Missing:** Add temperature/top-p as knobs that shape selection without overexplaining math.

ILLUSTRATION NEEDED
Current visual aid: sampling. Weighted vocabulary cloud with one selected token.



Sample One Token
Sampling chooses one token from the probability cloud.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 4/5	**Prompt/response** 5/5	**Durable/temp** 4/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

sampling · generated token · decoding step · response tokens · temperature · top-k · top-p

Autoregression

One token, append, repeat

Stage: response generation | **Priority: low** | **Rubric avg: 4.4/5**

DEFINITION
Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

CORE EXPLANATION
Autoregression means response generation by repeated next-token prediction: choose one token, append it, and run again.

WHERE IT HAPPENS
It is the loop that builds text during inference.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card belongs to decoding: the model is choosing one next response token, appending it, then repeating.

WHY IT MATTERS
Autoregression explains why earlier generated tokens become part of what the model can use next.

RELATIONSHIP
Each generated response token becomes part of the input context for the next decoding step.

METAPHOR
A train adding one car at a time.

BRAIN METAPHOR
It resembles continuing a thought step by step.

BRAIN LIMIT
The model is not holding a plan in mind; it repeatedly predicts the next token from visible context.

MISCONCEPTION
Learners may still imagine the model drafting a full paragraph internally before showing it.

VISUAL AID
`autoregression`

CURRENT EXERCISE
`prompt-or-response-label`: Prompt or Response?

CHECKPOINT
How does an LLM normally generate text? Correct: One token at a time. Incorrect: All words at once; By rewriting its weights.

FEEDBACK
LLM generation is autoregressive: next token, append, repeat.

REWRITE NOTES
Risk: Learners may still imagine the model drafting a full paragraph internally before showing it. **Missing:** Use one short dog/cat sentence to show next token, append, repeat.

ILLUSTRATION NEEDED
Current visual aid: autoregression. Loop diagram: score, sample, append, run again.

Append and Repeat
The chosen token is appended, then the model runs again.

Rubric Scores

Accuracy 5/5	**Placement** 5/5	**Relationship** 5/5	**Prompt/response** 5/5	**Durable/temp** 4/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 4/5	**Mobile** 4/5

autoregression · completion · generated token · response tokens · sampling · inference

Context Window

Temporary working context

Stage: current context | Priority: high | Rubric avg: 4.5/5

DEFINITION
A context window is the limited amount of tokens or media the model can currently use.

CORE EXPLANATION
A context window is the limited amount of tokens or media the model can currently use.

WHERE IT HAPPENS
It surrounds the prompt, conversation history, retrieved material, and response-so-far that remain visible.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card should separate the temporary visible context from durable memory or weight updates.

WHY IT MATTERS
It explains why models can use current context but do not automatically form permanent memory from every chat.

RELATIONSHIP
The model can only use tokens currently inside the context window, whether they came from the prompt, prior conversation, retrieved documents, or the response-so-far.

METAPHOR
A moving spotlight over a growing train.

BRAIN METAPHOR
It is loosely like working memory: what is currently available matters.

BRAIN LIMIT
It is not durable memory. What falls out cannot be directly attended to in the next pass.

MISCONCEPTION
Current context can be confused with saved memory or training data.

VISUAL AID
`context-window`

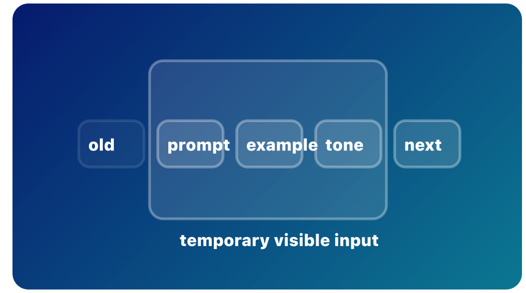
CURRENT EXERCISE
`context-window-fell-out`: Context Window: What Fell Out?

CHECKPOINT
What can happen when the context window is full? Correct: Older tokens may be truncated. Incorrect: The model permanently learns them; Softmax disappears.

FEEDBACK
Context is temporary input, not durable training.

REWRITE NOTES
Risk: Current context can be confused with saved memory or training data. **Missing:** Show what falls out and that retrieval adds text to context rather than training weights.

ILLUSTRATION NEEDED
Current visual aid: context-window. Limited stack of cards where older cards fall out.



Temporary Context Window

Only visible context can influence the next token.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

context window · input context · prompt tokens · response tokens · rag · autoregression · inference

RAG and Retrieval

Retrieved text joins the current context

Stage: current context | Priority: high | Rubric avg: 4.6/5

DEFINITION
Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's context before generating a response.

CORE EXPLANATION
Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's context before generating a response.

WHERE IT HAPPENS
It happens during inference, before or during response generation, and usually does not change model weights.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
Retrieved documents become prompt/context tokens. The answer is still generated as response tokens one at a time.

WHY IT MATTERS
RAG helps learners see that the model is not magically knowing private documents; retrieved snippets become temporary context.

RELATIONSHIP
RAG depends on the context window: retrieved text becomes prompt/context tokens, then the model still uses attention, hidden states, logits, softmax, and sampling to generate response tokens.

METAPHOR
Open-book AI: the model gets notes before answering.

BRAIN METAPHOR
It is a little like looking something up in notes before answering a question.

BRAIN LIMIT
A human can judge sources, remember what was read, and reason about trust in richer ways. The model still generates likely response tokens from retrieved context and learned weights.

MISCONCEPTION
RAG can be mistaken for permanent learning, full file access, web search by itself, fine-tuning, or a guarantee that answers are true.

VISUAL AID
`rag-retrieval`

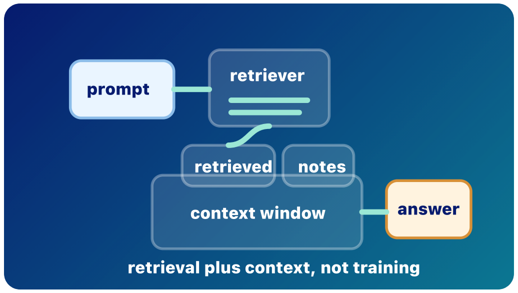
CURRENT EXERCISE
`open-book-or-learned`: Open Book or Learned?

CHECKPOINT
What does RAG usually do? Correct: Retrieves outside information and places it into the current context. Incorrect: Permanently updates the model's weights; Makes the model conscious of documents; Guarantees that every answer is true.

FEEDBACK
RAG is retrieval plus context. It can improve grounding, but it is not training, fine-tuning, permanent memory, or a truth guarantee. Choice notes: Permanently updates the model's weights: Not quite. Unless a separate training process happens, RAG does not durably update model weights. Makes the model conscious of documents: Not quite. Retrieved documents become visible context; the model does not become aware of them. Guarantees that every answer is true: Not quite. RAG can reduce hallucinations, but retrieval can be poor and the model can still misuse context.

REWRITE NOTES
Risk: RAG can be mistaken for permanent learning, full file access, web search by itself, fine-tuning, or a guarantee that answers are true. **Missing:** Name the retrieval step, the context insertion step, and the fact that weights remain unchanged unless training also happens.

ILLUSTRATION NEEDED
Current visual aid: rag-retrieval. Three lanes: user question, retriever searches a document shelf, retrieved cards enter the context window, and the LLM generates response tokens.



Open-Book Retrieval

Retrieved document cards enter the context before response tokens are generated.

Rubric Scores

Accuracy 5/5	Placement 5/5	Relationship 5/5	Prompt/response 5/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 5/5	Exercise 4/5	Mobile 4/5

rag · retrieval · grounding · context window · input context · prompt tokens · response tokens · inference · fine-tuning · training · hallucination

How AI Learns

Durable learning versus temporary steering

Stage: risk/policy | Priority: high | Rubric avg: 3.9/5

DEFINITION

AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.

CORE EXPLANATION

AI systems can improve or behave differently through durable training, targeted fine-tuning, retrieval, or temporary in-context steering.

WHERE IT HAPPENS

These processes happen at different times: before deployment, between deployments, or inside one current run.

DURABLE VS TEMPORARY

Not listed in current app data.

PROMPT VS RESPONSE

This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.

WHY IT MATTERS

Calling every behavior change learning creates confusion about privacy, memory, and institutional risk.

RELATIONSHIP

Pretraining and fine-tuning can durably change weights; retrieval and examples in the prompt change only what the model can see during inference.

METAPHOR

Revising the textbook versus bringing notes into an open-book exam.

BRAIN METAPHOR

The word learning is useful when prior experience changes later behavior.

BRAIN LIMIT

In-context examples are not personal learning; they steer the current run without normally updating weights.

MISCONCEPTION

Calling every behavior change learning can blur training, retrieval, and in-context steering.

VISUAL AID

`ai-learns`

CURRENT EXERCISE

None rendered in Journey; no direct Play mapping.

CHECKPOINT

Which process normally changes only the current context, not weights? Correct: Retrieval-augmented generation. Incorrect: Pretraining; Fine-tuning.

FEEDBACK

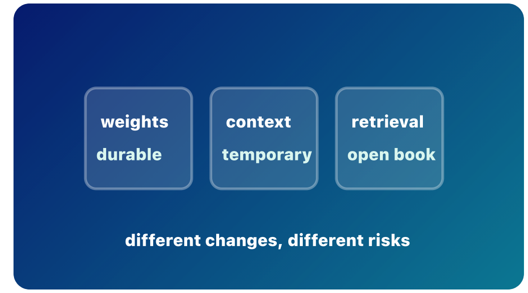
RAG places retrieved material into the current context. It does not normally train the base model.

REWRITE NOTES

Risk: Calling every behavior change learning can blur training, retrieval, and in-context steering. **Missing:** Separate durable training, retrieval, temporary instructions, and deployed online learning in one table.

ILLUSTRATION NEEDED

Current visual aid: ai-learns. Matrix of durable weight change, temporary context steering, and retrieval.



Learning Modes

Durable training, retrieval, and temporary steering change different things.

Rubric Scores

Accuracy 4/5	Placement 4/5	Relationship 4/5	Prompt/response 3/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 3/5

training · pretraining · fine-tuning · inference · rag · in-context learning

Diffusion vs Autoregression

Denoising is a different generation pattern

Stage: architecture | Priority: medium | Rubric avg: 3.6/5

DEFINITION
Diffusion generation usually starts from noise and denoises step by step.

CORE EXPLANATION
Diffusion generation usually starts from noise and denoises step by step.

WHERE IT HAPPENS
Diffusion is common in image generation and other media workflows, not the standard loop for text LLM generation.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
The comparison prevents learners from treating all generative AI as the same mechanism.

RELATIONSHIP
Autoregressive LLMs build text token by token; diffusion models refine noise into an image or other output.

METAPHOR
Writing the next word versus developing a photograph out of static.

BRAIN METAPHOR
Both systems can feel imaginative because they produce new outputs.

BRAIN LIMIT
The mechanisms differ: diffusion denoises a representation; autoregressive text predicts and appends tokens.

MISCONCEPTION
Learners may overgeneralize diffusion mechanics to text LLM generation.

VISUAL AID
`diffusion`

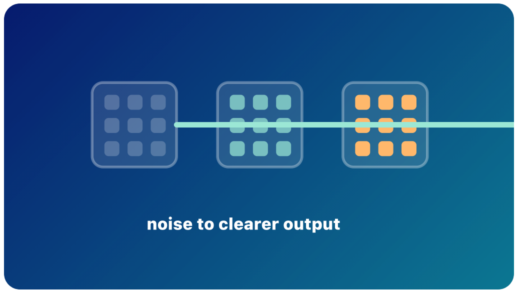
CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

CHECKPOINT
What does a diffusion model usually start with? Correct: Noise. Incorrect: A completed paragraph; A spreadsheet.

FEEDBACK
Diffusion models learn to denoise toward useful outputs.

REWRITE NOTES
Risk: Learners may overgeneralize diffusion mechanics to text LLM generation. **Missing:** Show a side-by-side: denoise image representation versus append text tokens.

ILLUSTRATION NEEDED
Current visual aid: diffusion. Denoising sequence beside a next-token text loop.



Denoise, Not Append
Diffusion refines noise step by step instead of generating text token by token.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 4/5	Prompt/response 2/5	Durable/temp 3/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 2/5	Mobile 4/5

diffusion | autoregression

Multimodal AI

Multiple media types together

Stage: architecture | Priority: medium | Rubric avg: 3.7/5

DEFINITION
Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

CORE EXPLANATION
Multimodal AI can represent or process multiple media types, such as text, images, audio, or video.

WHERE IT HAPPENS
It appears when systems connect encoders, decoders, or shared representations across media types.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card explains model machinery; it needs a visible bridge back to how a prompt becomes response tokens.

WHY IT MATTERS
It explains how a system can read an image, answer in text, or combine speech and documents.

RELATIONSHIP
Different modalities may use connected encoders, shared vector spaces, or coordinated model components.

METAPHOR
A transit hub where different kinds of information can transfer lines.

BRAIN METAPHOR
People combine sight, speech, and text naturally, so the metaphor helps.

BRAIN LIMIT
Machine modalities are engineered representations, not human sensory experience.

MISCONCEPTION
Human sensory metaphors can make engineered media representations sound more human-like than they are.

VISUAL AID
`multimodal`

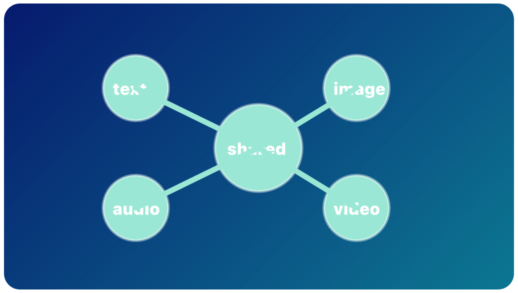
CURRENT EXERCISE
None rendered in Journey; no direct Play mapping.

CHECKPOINT
What does multimodal mean? Correct: Multiple media types. Incorrect: A larger context window only; A model with feelings.

FEEDBACK
Multimodal systems work across media types, often through learned representations.

REWRITE NOTES
Risk: Human sensory metaphors can make engineered media representations sound more human-like than they are. **Missing:** Add an example of image input, text output, and the representation bridge between them.

ILLUSTRATION NEEDED
Current visual aid: multimodal. Text, image, audio, and video paths meeting in a shared representation hub.



Shared Media Hub

Different media types can connect through learned representations.

Rubric Scores

Accuracy 5/5	**Placement** 4/5	**Relationship** 4/5	**Prompt/response** 3/5	**Durable/temp** 3/5
Metaphor 4/5	**Brain/cognition** 4/5	**Visual** 4/5	**Exercise** 2/5	**Mobile** 4/5

multimodal | embedding | vector

Risk vs Myth

Practical institutional literacy

Stage: risk/policy | Priority: high | Rubric avg: 4.3/5

DEFINITION
Risk literacy separates practical harms from magical stories about what models are.

CORE EXPLANATION
Risk literacy separates practical harms from magical stories about what models are.

WHERE IT HAPPENS
It matters whenever people use AI in classrooms, research, administration, health, advising, or IT systems.

DURABLE VS TEMPORARY
Not listed in current app data.

PROMPT VS RESPONSE
This card uses the prompt/response path to separate practical risk from myths about secret learning or agency.

WHY IT MATTERS
Clear mental models help institutions manage privacy, accuracy, bias, over-trust, and security without inventing magical threats.

RELATIONSHIP
Most real risks come from data, outputs, integrations, incentives, and human over-reliance; ordinary inference does not make the model secretly train itself.

METAPHOR
A campus safety map that marks real hazards clearly.

BRAIN METAPHOR
Brain metaphors can make risks feel urgent and memorable.

BRAIN LIMIT
Do not infer consciousness, intent, or secret self-training from fluent text. Keep the mechanism visible.

MISCONCEPTION
Fluent outputs can invite myths about consciousness, intent, or secret self-training.

VISUAL AID
`risk`

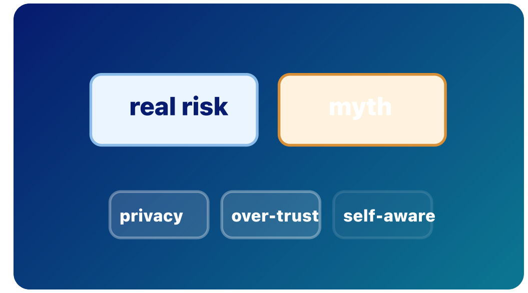
CURRENT EXERCISE
`open-book-or-learned`: Open Book or Learned?

CHECKPOINT
Which is a real institutional risk? Correct: Uploading private data. Incorrect: The model becoming conscious in the chat; Softmax stealing files.

FEEDBACK
Risk literacy is part of model literacy.

REWRITE NOTES
Risk: Fluent outputs can invite myths about consciousness, intent, or secret self-training. **Missing:** Tie each risk to a mechanism: context exposure, tool access, hallucination, bias, or over-trust.

ILLUSTRATION NEEDED
Current visual aid: risk. Risk sorting cards linked to mechanisms rather than fear language.



Risk or Myth
Clear mechanisms help separate practical risk from magical stories.

Rubric Scores

Accuracy 5/5	Placement 4/5	Relationship 5/5	Prompt/response 4/5	Durable/temp 5/5
Metaphor 4/5	Brain/cognition 4/5	Visual 4/5	Exercise 4/5	Mobile 4/5

inference · training · context window · privacy · hallucination · prompt-injection